

PROJECT MODULE: TASK 4 CONFIGURATION REPORT

Target Infrastructure: Ubuntu 24.04 LTS Instance on Microsoft Azure

1. Scope & System Objective

To implement strict directory-level isolation policies leveraging native POSIX security models. This ensures log directory paths follow the Principle of Least Privilege (PoLP) by restricting access exclusively to a dedicated identity profile.

2. Step-by-Step Security Hardening Execution

1. **Low-Privilege User Provisioning** : Created an isolated execution system context account:

Bash

```
sudo adduser monitor_user
```

2. **Container Engine Security Delegation** : Granted target workspace credentials access permissions to interface with background Docker engine components safely:

Bash

```
sudo usermod -aG docker monitor_user
```

3. **Volume Ownership Transfer** : Switched path permissions metadata parameters to align with the new service account identity:

Bash

```
sudo mkdir -p /opt/container-monitor/logs
```

```
sudo chown -R monitor_user:monitor_user /opt/container-monitor
```

4. **Access Masking Enforcement** : Stripped global visibility attributes across all standard directory scopes:

Bash

```
sudo chmod 700 /opt/container-monitor
```

```
sudo chmod 700 /opt/container-monitor/logs
```

3. Verification & Isolation Validation Metrics

- **Boundary Verification Test A: Non-Privileged Regular System Profiles (azureuser) :**

Plaintext

```
azureuser@prasad-vm:~$ cat /opt/container-monitor/logs/container_stats.log
```

```
cat: /opt/container-monitor/logs/container_stats.log: Permission denied
```

- **Boundary Verification Test B: Legitimate Service Target User (monitor_user) :**

Plaintext

```
monitor_user@prasad-vm:~$ cat /opt/container-monitor/logs/container_stats.log
```

[2026-06-18 15:22:01] Container: my-running-app | CPU: 0.01% | Mem: 2.635MiB / 7.756GiB

```
azureuser@prasad-vm: ~  
azureuser@prasad-vm:~$ cat /opt/container-monitor/logs/container_stats.log  
cat: /opt/container-monitor/logs/container_stats.log: Permission denied  
azureuser@prasad-vm:~$
```

```
monitor_user@prasad-vm: ~  
monitor_user@prasad-vm:~$ sudo su - monitor_user  
monitor_user@prasad-vm:~$ tail -n 10 /opt/container-monitor/logs/container_stats.log  
[2026-06-19 06:31:01] Container: my-running-app | CPU: 0.00% | Mem: 2.645MiB / 7.756GiB  
[2026-06-19 06:32:01] Container: my-running-app | CPU: 0.00% | Mem: 2.648MiB / 7.756GiB  
[2026-06-19 06:33:01] Container: my-running-app | CPU: 0.00% | Mem: 2.895MiB / 7.756GiB  
[2026-06-19 06:34:01] Container: my-running-app | CPU: 0.00% | Mem: 2.645MiB / 7.756GiB  
[2026-06-19 06:35:01] Container: my-running-app | CPU: 0.00% | Mem: 2.645MiB / 7.756GiB  
[2026-06-19 06:36:01] Container: my-running-app | CPU: 0.00% | Mem: 2.645MiB / 7.756GiB  
[2026-06-19 06:37:01] Container: my-running-app | CPU: 0.00% | Mem: 2.645MiB / 7.756GiB  
[2026-06-19 06:38:01] Container: my-running-app | CPU: 0.00% | Mem: 2.645MiB / 7.756GiB  
[2026-06-19 06:39:01] Container: my-running-app | CPU: 0.00% | Mem: 2.645MiB / 7.756GiB  
[2026-06-19 06:40:01] Container: my-running-app | CPU: 0.00% | Mem: 2.645MiB / 7.756GiB  
monitor_user@prasad-vm:~$
```